

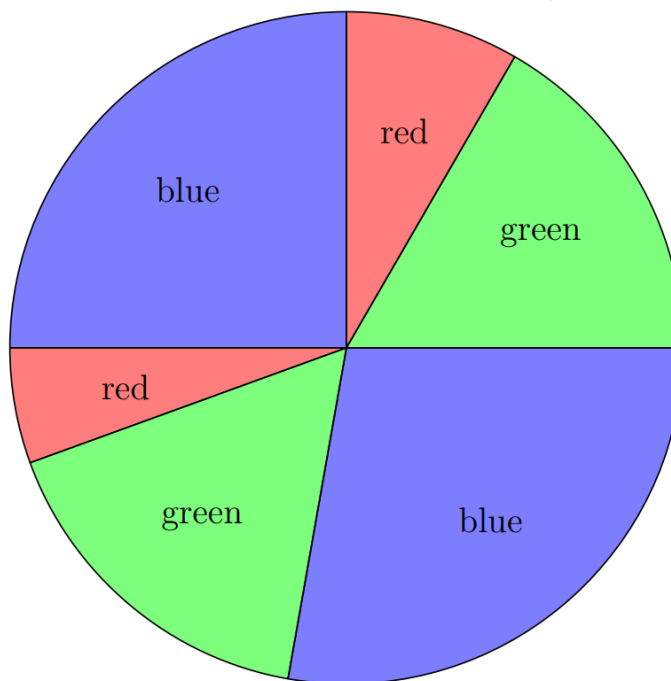
2025 AMC 10B Problem 16 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 16

Problem:

A circle has been divided into 6 sectors of 6 different sizes. Then 2 of the sectors are painted red, 2 painted green, and 2 painted blue so that no two neighboring sectors are painted the same color. One such coloring is shown below. How many different colorings are possible?



Answer Choices:

- A. 12
- B. 16
- C. 18
- D. 24
- E. 28



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Solution:

There are two cases on how the colors can be distributed:

Case 1: Every color is diametrically opposite itself.

For example if the sectors numbered in clockwise order are 1, 2, 3, 4, 5, 6, one color would be 1, 4, another would be 2, 5, and another would be 3, 6. There are $3!$ ways to assign the colors as such.

Case 2: Only 1 color is diametrically opposite itself.

There are 3 choices for the color and 3 choices for the sector pair for it to cover. Without loss suppose a color covered sectors 1 and 4. Then we see the two remaining colors must cover sectors 2, 6 and 3, 4. There are 2 ways to pick which color covers which sector for $3 \cdot 3 \cdot 2 = 18$ options in this case.

It is impossible to have exactly 0 or 2 diametrically opposite colors, so the answer is $18 + 6 = \boxed{\text{(D)} 24}$.

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